Simple Algebra

Year 4 · **Homework Worksheet**

Worked Example

Worked example: Find the missing number: 2 × □ + 3 = 11.

Answer: 2 × □ = 8, so □ = 4.

Easy / Standard (15 questions)

1. Find the missing number: 8 + □ = 15.
2. Find the missing number: □ − 5 = 11.
3. Find the missing number: 4 × □ = 24. A) 5 B) 6 C) 7 D) 8
4. Find the missing number: □ ÷ 4 = 6.
5. Find the missing number: 18 − □ = 5.
6. Select ALL of these where the missing number is 7: A) □ + 6 = 13 B) □ + 5 = 11 C) 3 × □ = 21 D) □ − 2 = 5 E) 20 ÷ □ = 4
7. Find the missing number: □ + 11 = 19.
8. Find the missing number: 7 × □ = 42. A) 5 B) 6 C) 7 D) 8
9. Find the missing number: □ ÷ 2 = 9.
10. Find the missing number: 16 + □ = 23.
11. Find the missing number: □ − 8 = 6.
12. Select ALL of these where the missing number is 5: A) □ × 4 = 20 B) □ + 9 = 14 C) 15 ÷ □ = 3 D) □ − 1 = 6 E) 12 + □ = 17
13. Find the missing number: 5 × □ = 35.
14. A function machine adds 3, then doubles. If the output is 22, what was the input? A) 7 B) 8 C) 9 D) 10
15. Find the missing number: □ + 12 = 25.

Standard (15 questions)

1. A function machine adds 3, then multiplies by 2. If the input is 5, what is the output?
2. A function machine subtracts 4, then triples. If the input is 9, what is the output?
3. A function machine multiplies by 3, then adds 4. If the input is 5, what is the output? A) 15 B) 17 C) 19 D) 21
4. Find the missing number: 3 × □ + 5 = 20.
5. Find the missing number: □ ÷ 3 + 2 = 8.
6. Select ALL of these where the missing number is 6: A) 2 × □ = 12 B) □ + 7 = 13 C) 30 ÷ □ = 6 D) □ − 3 = 9 E) 3 × □ − 2 = 16
7. Find the missing number: 4 × □ − 5 = 19.
8. A function machine adds 6, then halves. If the input is 8, what is the output?
9. Find the missing number: 5 × □ + 3 = 33. A) 5 B) 6 C) 7 D) 8

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10. Find the missing number: □ ÷ 5 − 1 = 3.

11. A function machine doubles, then adds 7. If the output is 23, what was the input?

12. Find the missing number: 6 × □ − 4 = 26.

13. Select ALL of these where the missing number is 8: A) □ + 6 = 14 B) 4 × □ = 32 C) 2 × □ − 3 = 13 D) 24 ÷ □ = 3 E) □ − 5 = 2

14. Find the missing number: 7 × □ + 2 = 44.

15. A function machine adds 5, then triples. If the output is 27, what was the input? A) 3 B) 4 C) 5 D) 6

Standard / Advanced (15 questions)

1. Find the missing number: (□ + 2) × 2 = 14. A) 4 B) 5 C) 6 D) 7

2. I think of a number, subtract 3, then multiply by 2. The answer is 10. What was the number?

3. Select ALL of these where the missing number is 7: A) 2 × □ − 3 = 11 B) □ + 5 = 12 C) 2 × (□ − 4) = 6 D) 35 ÷ □ = 5 E) 3 × □ − 1 = 23

4. Find the missing number: 4 × □ − 2 × □ = 18.

5. A farmer has chickens and cows. Chickens have 2 legs, cows have 4 legs. There are 5 animals with 14 legs in total. How many cows are there? A) 1 B) 2 C) 3 D) 4

6. Two numbers add to 20. One number is 3 times the other. What are the two numbers?

7. A rectangle's length is 3cm more than its width. The perimeter is 22cm. Find the width.

8. A shop sells notebooks for £2 each and pens for £1 each. Josh buys some notebooks and 2 pens, spending £10 in total. How many notebooks did he buy? A) 2 B) 3 C) 4 D) 5

9. A ribbon is cut into two pieces. One piece is 3 times as long as the other. If the ribbon is 40cm long, how long is the shorter piece?

10. Find the missing number: 2 × □ + 3 × □ = 25.

11. A taxi charges a fixed £2 plus £3 per mile. A journey costs £14. How many miles was the journey?

12. Sam is 4 years older than his sister. In total their ages add up to 20. How old is Sam?

13. A box contains red and blue counters, with twice as many red as blue. Select ALL of the total counts below for which this split could work with a whole number of blue counters: A) 9 B) 10 C) 12 D) 14 E) 15

14. A school buys pencils at 10p each and spends a total of £2.00. How many pencils did they buy?

15. A function machine adds 3, then doubles the result. If the input is 5, what is the output?

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